

*Haven.*

MIDNIGHT PREPROD

• SYS.ONLINE

• MIDNIGHT NETWORK • ZERO-KNOWLEDGE ESCROW

# *Private, Zero-Knowledge Escrow on Midnight.*

Real shielded token custody. Zero-knowledge proofs for every state transition. Nothing sensitive is ever exposed on-chain.

**Anubhab Rakshit**

Midnight Buildathon 2026



## THE PROBLEM

# *Traditional escrow is broken.*

Buyers and sellers expose amounts, addresses, and conditions publicly on-chain — and anyone can analyze the transaction graph to deanonymize participants.



### *Transparent amounts*

Competitors, bots, and MEV actors see every deposit, release, and price in the clear.



### *Identity leakage*

Wallet addresses link directly to real-world identities through public transaction history.



### *Fake privacy*

Most "private" escrow dApps simulate shielded tokens — they never touch a real ZK circuit.

## THE SOLUTION

# *Haven: escrow where nothing leaks.*



### *Real shielded tokens*

Deposits mint actual ZK coins on-chain via Compact's mintShieldedToken — not a mock balance.



### *Zero-knowledge proofs*

Every state transition — deposit, release, cancel — generates and verifies its own ZK proof.



### *Domain-separated commitments*

Buyer, seller, amount, and condition are each hashed with unique prefixes to block cross-domain attacks.



### *Observer-blind ledger*

Public state exposes only counters and commitment hashes — amounts and identities stay hidden.

BUYER → SHIELDED CONTRACT → SELLER



## HOW IT WORKS

# System architecture, in three layers.



## STATE MACHINE



+ dispute / cancel branches at Funded and Delivered

1

### Create escrow

Buyer, seller, amount, and condition are hashed into commitments and written on-chain.

2

### Buyer deposits

mintShieldedToken mints a ZK coin; getCoinMtlIndex indexes it into the tree.

3

### Seller confirms delivery

A state transition fires with its own zero-knowledge proof of validity.

4

### Buyer releases

sendShielded transfers the coin to the seller's shielded key — done.

# *Built on Midnight's privacy-first infrastructure.*



## Smart Contract

Compact 0.23 — Midnight's ZK language



## Frontend

React 19 · Vite 6 · Framer Motion · GSAP · Lenis



## Wallet

Midnight Wallet SDK + Lace browser extension



## Proof Server

midnightntwrk/proof-server on Render



## Backend

TypeScript · Express 5 · Node 22



## Database

Supabase (PostgreSQL)



## Network

Midnight preprod testnet



## Deployment

Render (API + proof server) · Vercel (frontend)

# *What observers actually see vs. reality.*



## *Public ledger — what's visible*

- Commitment hashes
- State counters
- State enum (Created / Funded / ...)



## *Private — hidden by ZK*

- Buyer & seller identity
- Escrow amount & coin value
- Condition text
- Full transaction graph



## **Domain-separated commitments**

Every circuit hashes with persistentHash under a unique prefix — "midnight-lock:v1:buyer:", "midnight-lock:v1:seller:", and so on — to prevent cross-domain attacks.

*The deposit circuit mints shielded tokens with no unshielded intermediary. Release uses sendShielded — only the recipient can decrypt.*

# Production-ready infrastructure.

90

TESTS PASSING

contract · service · witnesses · e2e · private state

3

NETWORKS SUPPORTED

undeployed · preview · preprod

VERIFIED ON PREPROD

deploy → deposit(coinIndex=21232) → confirmDelivery → release

Environment-agnostic via MIDNIGHT\_NETWORK — local Docker devnet, preview, and preprod are all supported. Wallet seed management works via env vars or BIP-39 mnemonic backup.

Render deployment runs the API server and proof server together on an auto-scaling starter plan.



Live escrow dashboard — Haven on Midnight preprod

# *Where Haven goes next.*

## IMMEDIATE

---



Mainnet deployment when Midnight launches



Multi-party escrow (3+ participants)



Time-locked auto-release

## MEDIUM-TERM

---



Dispute resolution via decentralized arbiter DAO



Cross-chain escrow (BTC, ETH → Midnight shielded tokens)



Mobile-first PWA with biometric auth

## LONG-TERM VISION

---



Marketplace of escrow templates — rental, freelance, P2P



Privacy-preserving reputation — prove completion without revealing which



Integration with Midnight's stablecoin for real-world commerce